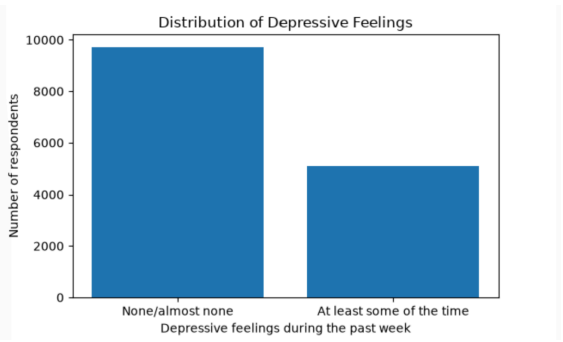
Nazanin Fallah Taija-Sophie Schaffer, Era Rexhepi

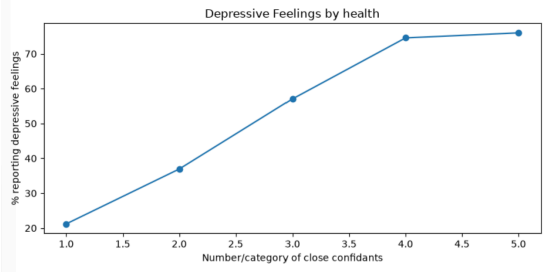
technical report

For our mini-project we wanted to make something that was simple and easy to use and could give a great benefit to a person's life. In these digital times, where news and misinformation are hard to escape and where good resources are hard to come by, we chose to create a website that would help people interested in finding resources about mental wellbeing and health.
We did not want to diagnose or take over any job that a therapist/psychiatrist would, we wanted to assess risk and give country specific resources to help people out on their journey of mental wellbeing.
But instead of just creating a website with the resources we wanted to find a way to personalize the information to lower the barrier of learning and getting help. Our target group was set to be Gen Z and young adults, or people interested in supporting someone from that age group.

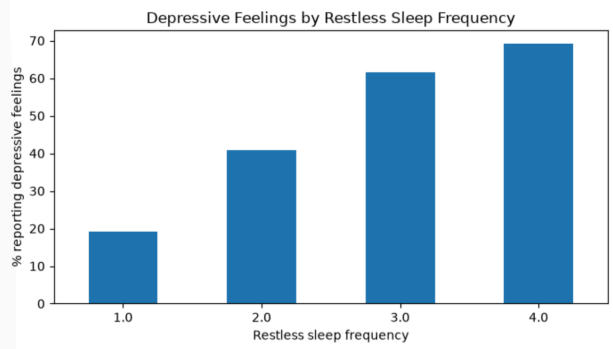
For that we used the **European Social Survey** dataset. This dataset contained 159.320 observations with 35 distinct features. Our first step was to match the data with our target group, using only adults from the generation of Millennial and Gen Z. Then we chose the variables that we wanted to examine: age, generation, country, internet use (daily), restlessness of sleep, perceived control over life, feeling depressed, feeling happy, general physical health, feeling lonely, feeling sad, gender, unemployed for over 3 months, amount of people subjects can discuss intimate and personal matters with
In the survey they used numbers like 6666,7777,etc. to represent different kinds of missing values. Since the survey was not done through a simple quiz, we decided to simplify the data by just replacing all those different missing variables with nan.
For our variable of interest (feelings of depression) we decided to fill all missing values with the mean.
Since the survey data contains multiple surveys done in different years, we need to select one that has our variable of interest populated 100%.
To check how many fully populated data points we have we check the percentages of Nan in all of our columns. The percentage of missing values for the internet usage per day was at 5.56%, therefore we decided to impute the missing values by their generation median in order to not lose too many datapoints. We found the other columns to have a Nan percentage between 0.27% and 0%. Therefore we decided to drop these, since the loss in this case is negligible.
At this point we could report that our data is made up of 73% Millennial and 27% Gen Z correspondents. As well as that 53% of respondents were women and 47% were men.

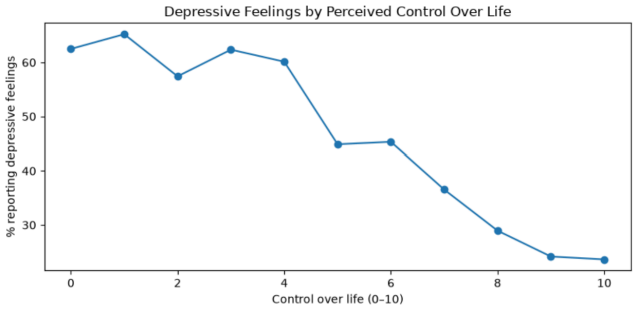


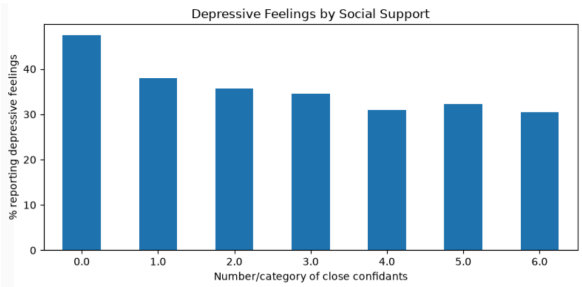
We decided to plot some more variables to get a deeper understanding of our data.
We found that the group reporting the highest average score (1.51) of depressive feelings was Gen Z with high internet usage (6+h), however for Millenials the highest (1.45) was found in the group of 3-6h of internet usage.
We found that overall almost 30% of participants in our dataset reported to have depressive



feelings at least some of the time. Therefore we decided to create some plots looking at possible correlations.
In this analysis we found that perception of health, the restlessness of sleep and the perceived control over one's life seem to be important to the reporting of depressive feelings.







It was however surprising that the amount of people that subjects could discuss intimate and personal matters with did not seem to affect the reporting of depressive feelings very much. While no social contacts still score the highest numbers of reported depression, even people with a high number of contacts still report depressive feelings at 30%.

This concluded our visualization and data analysis and we moved onto the modeling part. For our modeling we decided to test a Logistic Regression model, as well as a Random Forest model.

	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.698	0.550	0.664	0.602	0.759
Random Forest	0.697	0.545	0.706	0.615	0.765

Neither model clearly dominated. While Logistic Regression achieved a slightly higher precision, Random Forest achieved higher recall. Making this model better at identifying positive cases.
We decided to go with the Random Forest model, since it performed slightly better overall, even with a minorly lower score at precision.
The Random Forest classifier was trained using 300 decision trees (n_estimators = 300) with a maximum tree depth of 8 (max_depth =8). This was chosen to reduce the chance of overfitting while allowing the model to capture non-linear relationships.

In our feature importance we found the restlessness of sleep (0.333), perception of health(0.204), as well as the perceived control over one's life (0.143) to be the main features. These are soon followed by time spent on the internet (0.049) and the amount of people the subject can discuss personal and intimate matters with (0.417). These results were not surprising considering the findings we had made in our EDA and correlation plots.
We gather the information needed to predict the likelihood of reporting depressive feelings through a quiz, which we then map to the correct variables.
In our quiz we give the option to choose from more countries than those represented in our data. This is so we can still give country specific resources, even if we don't have any available data from the country of the user.
For the individual result, we take the predicted chance of reporting depression, as well as looking at the most important features. Depending on their answers, we will let them know if the scores on those features might influence their reported depressive feelings. Those are not treatments. We simply use those main features to let them know that these might be affecting their reported depressive feelings. With all personalized results also come general resources on mental health and links to international foundations that specialize in mental health in youth and young adults.
If our prediction is very high, we also include country specific emergency hotlines and encourage users to reach out for help.

Our first idea for the workflow was to create an informative website, explaining the importance of mental health resources and the increase of mental illnesses in younger generations. Followed by a link to a quick quiz, where we would collect some of the most important information to predict the risk of the participant to report depressive feelings often. Then we wanted to send out a personalized email with country specific resources.
Because our project involves sensitive information related to personal well-being, we wanted to handle all data with the utmost care and prioritize our users' privacy. To minimize the possibility of linking the quiz responses to any individual, we decided against personalized emails. Instead, the results and relevant resources will still be personalized, but instead displayed directly on the Website. This way we can ensure that users can access all information in one place while reducing the need to store personal data anywhere.